

University of Applied Sciences Hamburg	DEPARTMENT OF ELECTRICAL ENGINEERING	Wheatstone Bridge calibration
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Report History

Report 1		Date	Remarks
	received		
	checked		
	result	o.k.	
		n.g.	→ 1. Correction → Term.....
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		n.g.	→ 2. Correction → Term.....
2. Correction		Date	Remarks
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	result	o.k.	
		n.g.	→ not passed → back to L1.....

Final decision:

☐ o.k.

☐ not passed

Prof.

Abstract

In this lab, we implemented a strain gauge-based measurement system using a Wheatstone Bridge configuration and National Instruments LabVIEW software for real-time data acquisition and processing. The experiment began with setting up a bending beam equipped with four strain gauges in a full-bridge arrangement, powered by a 6V supply. Measurements were taken using both a multimeter and an oscilloscope, allowing for a comparison of observed voltage values and understanding measurement discrepancies caused by instrument response. Calibration of the Wheatstone Bridge was achieved using a resistor decade to balance the circuit and fine-tune sensitivity to $E_0 = 1\text{mV}/100\text{g}$. Following hardware calibration, an NI-DAQ device was connected to the system to enable differential voltage measurements via LabVIEW. To ensure system accuracy, software calibration was implemented to compensate for changes in supply voltage and hardware adjustments. The linear behavior of the bridge voltage with respect to weight changes was verified, confirming the system's suitability for precision measurement applications. This lab integrated hardware setup, data acquisition, and software processing, providing hands-on experience in designing and calibrating sensor-based measurement systems using LabVIEW.

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1 Hardware

1.1 Set-Up

The experiment involves the implementation of a **Wheatstone Bridge** circuit using four strain gauges on a bending beam to measure deformation under load. The hardware setup ensures accurate voltage readings from the bridge circuit through multimeter and oscilloscope measurements.

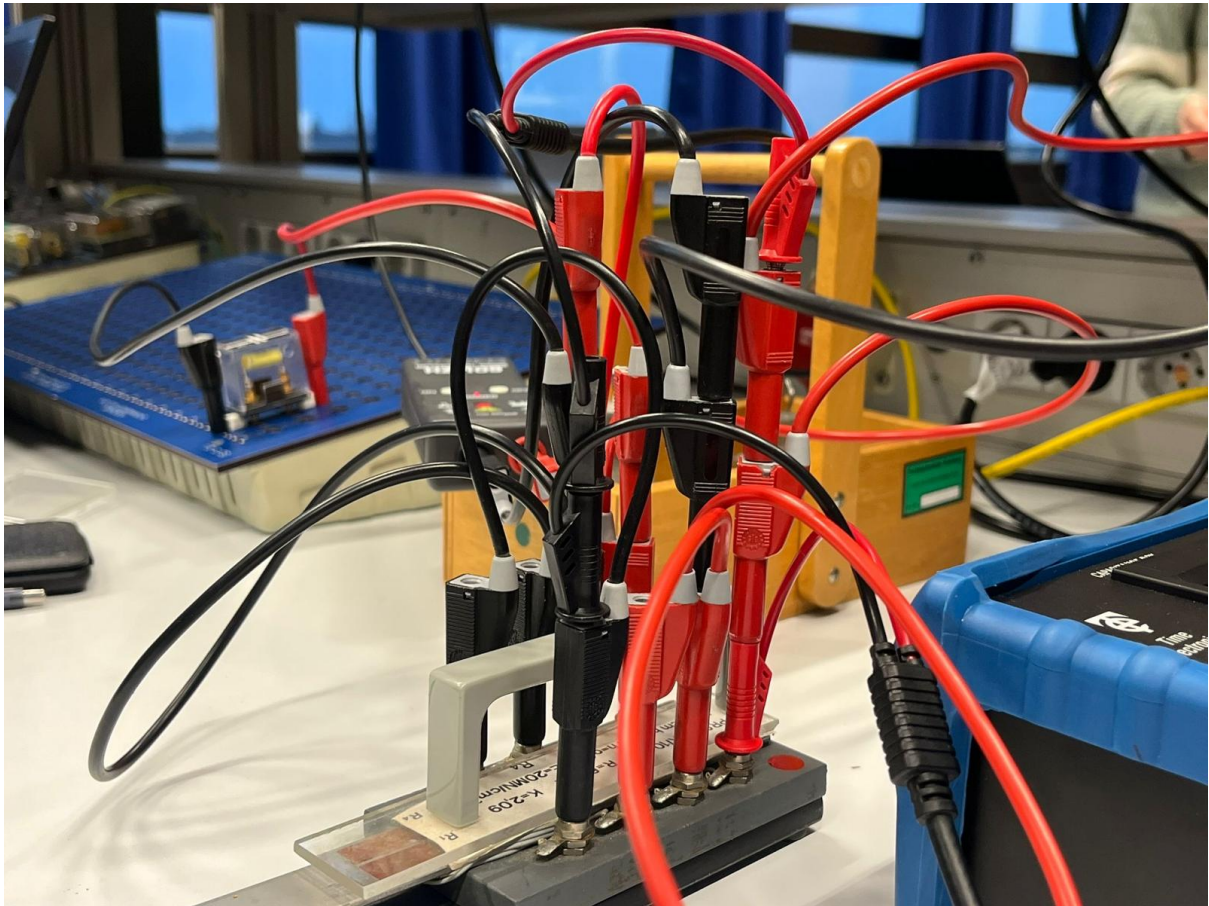


Figure 1: Circuit set-up

Observations:

- The **multimeter** provides a steady DC voltage reading corresponding to the bridge imbalance caused by strain on the beam.
- The **oscilloscope** displays a time-domain waveform of the bridge voltage. Minor noise or ripples might be observed due to external interference or supply variations.

1.2 Calibration

Balancing The Bridge:

- Replace the resistor decade with an adjustable capacitor in parallel to one of the strain gauges.
- Gradually tune the capacitor to balance the bridge, reducing the output voltage to near zero.
- Use the sensor to apply a known load and observe how the output voltage changes as the capacitor is adjusted.

Adjusting sensitivity:

- Calibrate the bridge such that the output voltage meets the sensitivity requirements :

$$E_0 = 1 \text{ mV}/100 \text{ g}$$

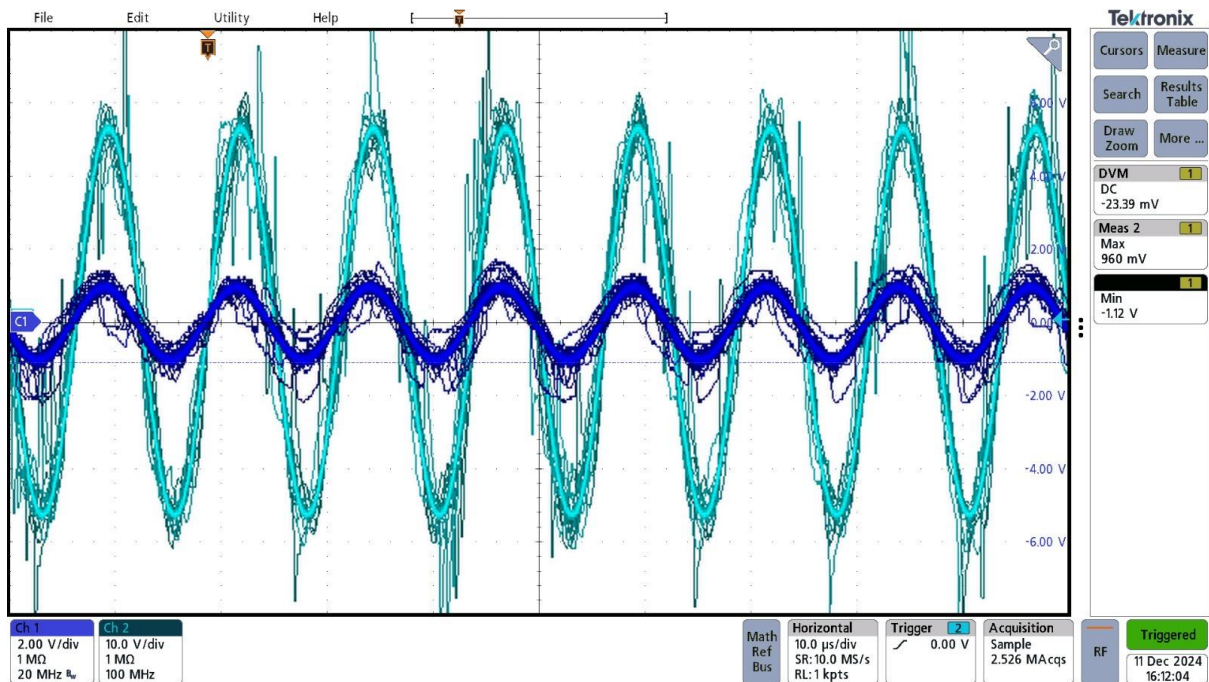
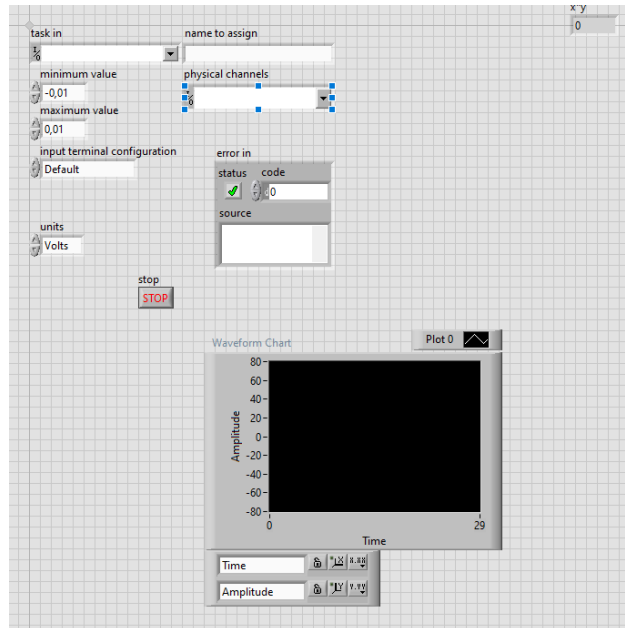


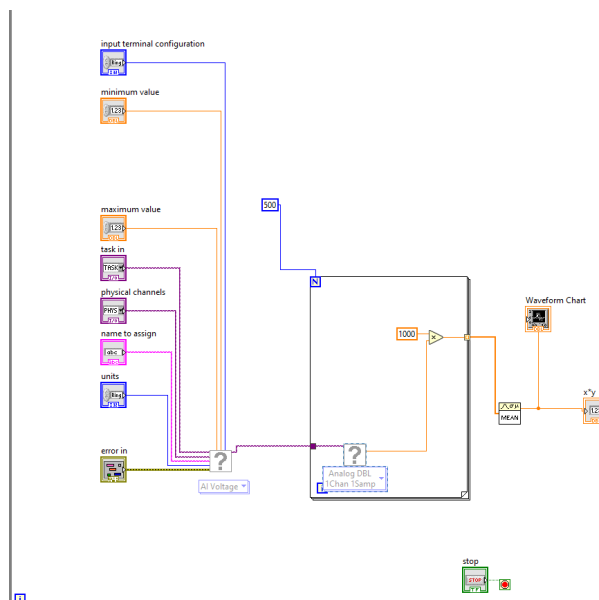
Figure 2: Balanced Bridge Diagram

2 Data Acquisition

The data acquisition process involves capturing the output voltage of the Wheatstone Bridge using a (NI) DAQ device and processing the data in LabVIEW. This step ensures that the measured bridge voltage is accurately converted into corresponding weight values. The process includes hardware connections to the NI-DAQ device, configuring the LabVIEW software for continuous voltage measurement, and observing the real-time data displayed as both numerical values and graphical indicators. The acquired data is further calibrated and scaled to ensure accurate representation of the applied loads.



(a) Front Panel



(b) Block Diagram

Figure 3: DAQ read basic v

3 Observations

3.1 Output at 6V:

The waveform displayed minor fluctuations around a central value, indicating initial imbalances in the bridge circuit due to noise or small misalignments.

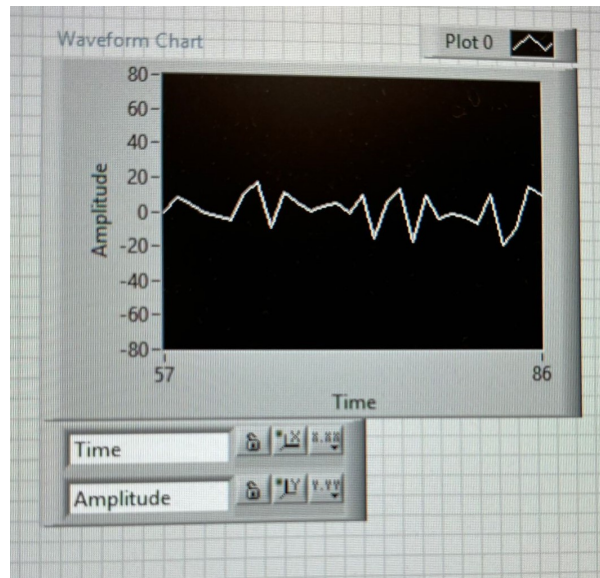


Figure 4: output at 6V

3.2 Output at 4.5V:

The waveform remained stable with reduced amplitude, confirming the bridge output's linear dependence on the supply voltage.

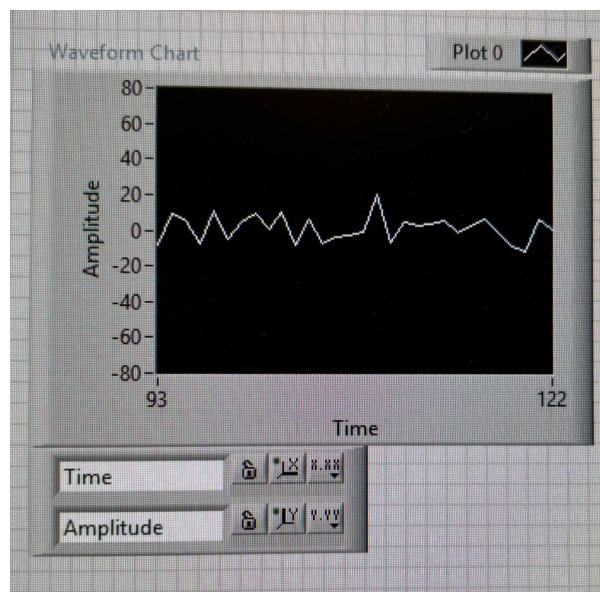


Figure 5: output at 4.5V

3.3 Output at 6V with 0.2kg load:

The waveform exhibited a noticeable shift in amplitude, corresponding to the load-induced strain measured by the strain gauges. This verifies the bridge circuit's ability to detect weight variations accurately.

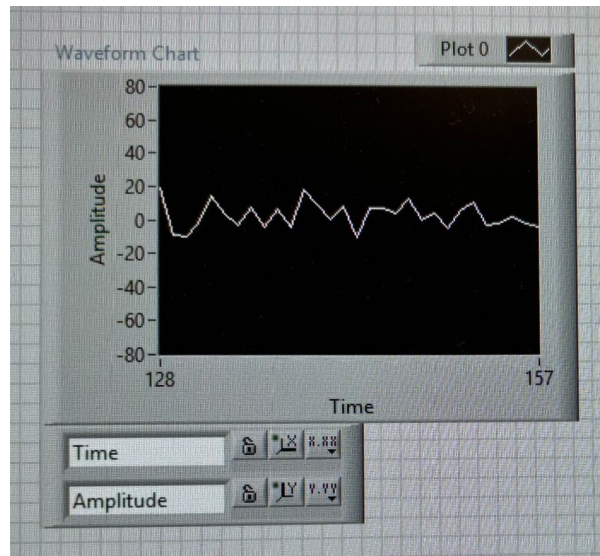


Figure 6: output at 6V with 0.2kg load